

Ex: 1007

16-09-25 BUILD A CNN MODEL TO CLASSIFY CAT & DOG IMAGES

Aim:

To build and train CNN model that classifies images into 2 categories cat & dog

Objectives:

1. To understand the architecture of CNN
2. To preprocess image dataset for training and testing
3. To implement a CNN model using PyTorch
4. To evaluate the performance (model's accuracy & loss).

Pseudocode:

1. Import necessary libraries (Torch, Torch version)
 2. Load the dataset
 3. Create a clean CNN with
 - > convolutional layer + ReLU + max pooling
 - > fully connected layer
 - > output layer with 2 neurons
 4. Train the data
- For each epoch:

For each batch in training data.

Forward pass

compute loss

Backpropagate loss

update weights

print training loss

3. Testing:

Evaluate model on Test data

Calculate accuracy

END.

Observation:

1. Dataset.

Dataset contains image of cats and dogs.

Image resized to 128×128 normalized before training.

2. At beginning of training, CNN had randomly initialized weights.

3. As multiple epoch, CNN layer begin to extract meaning full features such as edges, textures, shape.

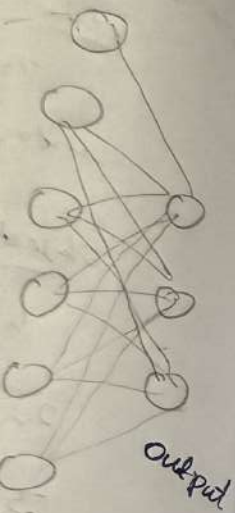
4. pooling layer helped reduce dimensionality.

5. After several epoch, model loss decreases while accuracy improves.

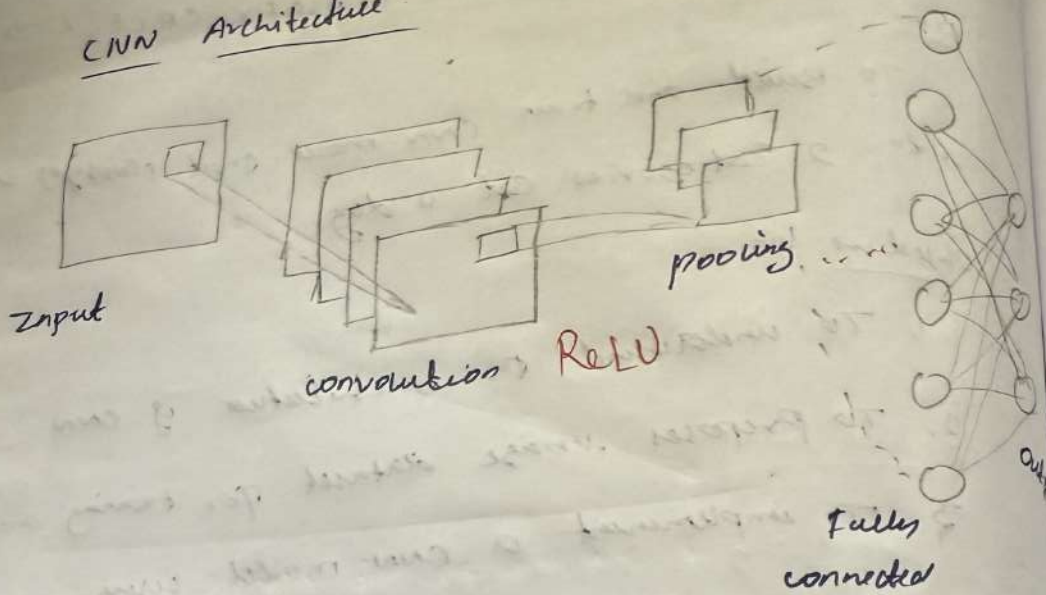
Result:

implemented - Build CNN model to classify cat & dog image.

val	acc
0.56	
0.60	
0.64	
0.68	
0.72	



CNN Architecture



Output: epoch	Loss	Accuracy	val acc
epoch 1/5	Loss: 0.6820	0.55	0.56
epoch 2/5	Loss: 0.6375	0.62	0.60
epoch 3/5	Loss: 0.5990	0.67	0.64
epoch 4/5	Loss: 0.5400	0.73	0.68
epoch 5/5	Loss: 0.5011	0.77	0.72

Test Loss: 0.4952

Test Accuracy: 0.7350

3. Testing:

Evaluate

Calculate

END.

Observation

1. Data

Dataset

Image

Training

2.

Initial

3.

mean

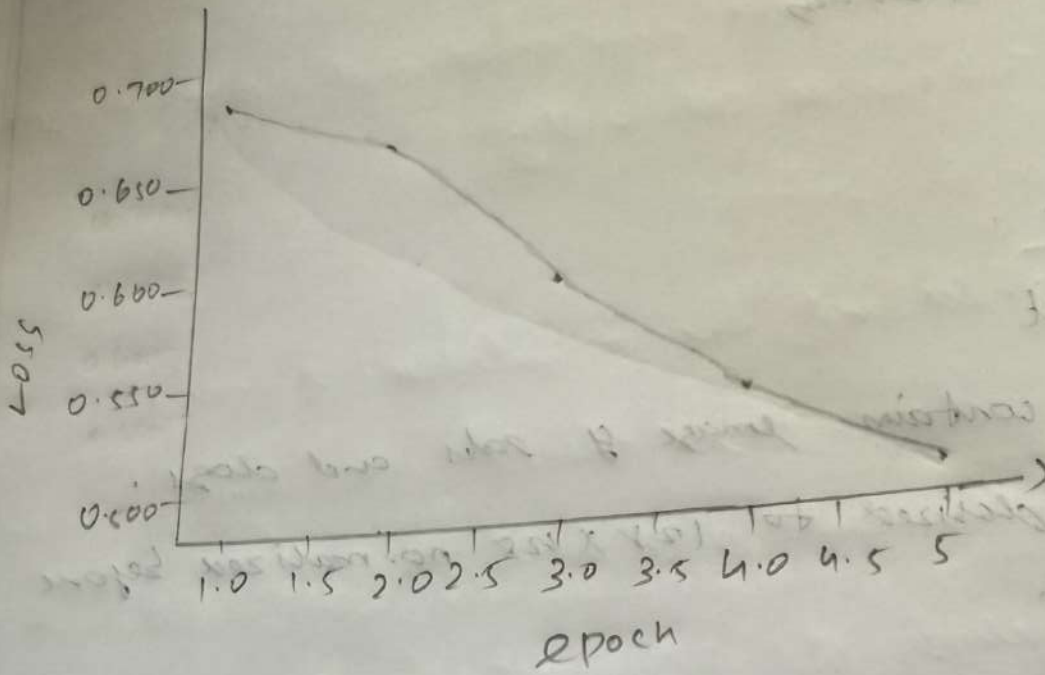
Shape

4.

5.

with

Loss over



Result

~~Result~~

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Untitled.py...	14 days ago

lab7.ipynb

```
[1]: import matplotlib.pyplot as plt

history = {
    'loss': [0.6820, 0.6375, 0.5990, 0.5400, 0.5011]
}

epochs = range(1, 6)

# Plotting loss over epochs
plt.figure(figsize=(8,5))
plt.plot(epochs, history['loss'], label='Training Loss', color='blue', marker='o')
plt.title('Loss Over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()

plt.show()
```

Loss Over Epochs

Epochs	Loss
1	0.6820
2	0.6375
3	0.5990
4	0.5400
5	0.5011

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